Matrix discrepancy and optimization

The Komlós discrepancy conjecture

Difficulty: extreme **Importance:** broadly interesting

Provenance: source-stated conjecture.

Status: no resolution found in screening on 2026-09-08.

Does a finite constant $C > 0$ exist such that, for every pair of positive integers m, n and every matrix $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ satisfying

$$\sum_{i=1}^{m} a_{ij}^2 \le 1 \qquad (1 \le j \le n),$$

there is a vector $x \in \{-1, 1\}^n$ for which

$$\|Ax\|_\infty = \max_{1\le i\le m} \left| \sum_{j=1}^{n} a_{ij} x_j \right| \le C?$$

The same constant must work for every dimension and matrix. The question asks for existence of a signing; it does not additionally require an algorithm.

This is a matrix balancing problem: choose column signs so that their sum is small in every coordinate. Its norm constraint and simultaneous coordinate control connect discrepancy theory with rounding and linear optimization.

References

1. N. Bansal and H. Jiang, *An Exposition of the $\widetilde{O}(\log^{1/4} n)$ Bound for the Komlós Problem*, arXiv:2608.28452 (2026), abstract and introduction, including the explicit conjecture and the new dimension-dependent bound. Paper.
2. N. Bansal and H. Jiang, *Decoupling via Affine Spectral-Independence: Beck-Fiala and Komlós Bounds Beyond Banaszczyk*, arXiv:2508.03961v2 (2025), abstract and introductory results. Paper.
3. W. Banaszczyk, *Balancing vectors and Gaussian measures of n-dimensional convex bodies*, Random Structures & Algorithms 12(4) (1998), pp. 351–360, the main vector-balancing theorem underlying the earlier logarithmic bound. Article.

Status check — 2026-09-08

checked the current arXiv records (2608.28452v1, 2026-08-28; 2508.03961v2, 2025-09-09) and searched "Komlos conjecture solved proof September 2026" and "Komlos Bansal Jiang 2026 bound". The new bound is $O((\log n)^{1/4}(\log\log n)^{7/4})$ in its asymptotic range, not a universal constant. No full proof, counterexample, or withdrawal of these source papers was found. The disproved Hajela conjecture and the separate matrix Spencer conjecture are not this statement.